

Least Squares

Theory Questions and Solutions

TTK4260

Question 1 (*Statistical Foundations*)

Consider a dataset with observations $\{y_1, y_2, \dots, y_n\}$.

- (a) Define the sample mean, sample variance, and explain their roles in least squares estimation. [6 points]
- (b) Explain why we square the residuals instead of using absolute values in the least squares formulation. [4 points]

Solution.

- (a) The **sample mean** is

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i,$$

and the **sample variance** is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2.$$

The sample mean minimizes the sum of squared deviations from a constant model and is therefore a least squares estimator. The sample variance quantifies the spread of the data and reflects the expected magnitude of residuals in least squares estimation.

- (b) The fundamental assumption is that the measurement errors ε_i are independent, zero-mean random variables with constant variance. Under Gaussian noise, least squares corresponds to maximum likelihood estimation.
- (c) Squaring residuals ensures differentiability, penalizes large errors more heavily, and leads to a convex optimization problem with a closed-form solution. Absolute values lead to non-differentiability at zero and require more complex optimization methods.

Question 2 (*Least Squares Problem Formulation*)

Consider the linear regression model $y_i = \theta_0 + \theta_1 x_i + \varepsilon_i$ for $i = 1, 2, \dots, n$.

- (a) Write the cost function $J(\boldsymbol{\theta})$ that least squares minimizes. [3 points]
- (b) Express this problem in matrix form $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$ and define each component. [6 points]
- (c) What is the geometric interpretation of the least squares solution? [5 points]

Solution.

- (a) The least squares cost function is

$$J(\boldsymbol{\theta}) = \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i)^2.$$

(b) In matrix form:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix},$$

giving

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}.$$

(c) Geometrically, the least squares solution is the orthogonal projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$. The residual vector is orthogonal to this space.

Question 3 (*Normal Equations Derivation*)

Derive the normal equations for the least squares problem from first principles.

- (a) Start with the objective function $J(\boldsymbol{\theta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2$ and expand it fully. [6 points]
- (b) Take the gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$ and set it equal to zero. [8 points]
- (c) Derive the closed-form solution $\hat{\boldsymbol{\theta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$. [6 points]
- (d) Under what conditions does this solution exist and is it unique? [4 points]

Solution.

(a) Expanding the objective:

$$J(\boldsymbol{\theta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) = \mathbf{y}^T \mathbf{y} - 2\boldsymbol{\theta}^T \mathbf{X}^T \mathbf{y} + \boldsymbol{\theta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\theta}.$$

(b) Taking the gradient:

$$\nabla_{\boldsymbol{\theta}} J = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \boldsymbol{\theta}.$$

Setting this equal to zero yields the normal equations.

(c) Solving:

$$\mathbf{X}^T \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^T \mathbf{y} \quad \Rightarrow \quad \hat{\boldsymbol{\theta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.$$

(d) The solution exists and is unique if $\mathbf{X}^T \mathbf{X}$ is invertible, which requires the columns of $\mathbf{X}$ to be linearly independent.

Question 4 (*Gradient Descent Theory*)

Consider using gradient descent to solve a least squares problem.

- (a) Write the gradient descent update rule for the parameter vector $\boldsymbol{\theta}$. [4 points]
- (b) Explain the difference between batch gradient descent and stochastic gradient descent. [6 points]
- (c) Why is feature scaling important for gradient descent? Provide a mathematical explanation. [6 points]
- (d) What factors affect the convergence rate of gradient descent in least squares problems? [4 points]

Solution.

- (a) The update rule is

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - \alpha \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}^{(k)}),$$

where α is the learning rate.

- (b) Batch gradient descent uses all data points per update, while stochastic gradient descent updates parameters using one (or a small batch of) data point(s), leading to faster but noisier convergence.
- (c) Feature scaling improves conditioning of $\mathbf{X}^T \mathbf{X}$, reducing the spread of eigenvalues and allowing gradient descent to converge faster and more stably.
- (d) Convergence depends on the learning rate, conditioning of the Hessian, feature scaling, and the choice of initialization.

Question 5 (*Optimization Methods Comparison*)

Compare different approaches for solving least squares problems.

- (a) Describe the brute force (grid search) approach. When might this be useful despite its computational cost? *[5 points]*
- (b) Compare the advantages and disadvantages of analytical solutions (normal equations) versus iterative methods (gradient descent). *[8 points]*

Solution.

- (a) Grid search evaluates the cost function over a discretized parameter space. It is useful for low-dimensional problems, visualization, or validation of other optimization methods.
- (b) Normal equations provide exact solutions but scale poorly with large feature sets and may be numerically unstable. Gradient descent scales better to large datasets but requires careful tuning and convergence monitoring.

Question 6 (*Non-linear Least Squares*)

Consider a model that is non-linear in its parameters: $f(x, \boldsymbol{\theta}) = \theta_1(1 - e^{-\theta_2 x})$.

- (a) What optimization methods are suitable for non-linear least squares? *[6 points]*
- (b) How would you implement parameter estimation for this model using gradient-based methods? *[7 points]*
- (c) What challenges might arise during optimization and how would you address them? *[6 points]*

Solution.

- (a) Suitable methods include gradient descent, Gauss–Newton, Levenberg–Marquardt, and trust-region methods.

- (b) One defines the residuals

$$r_i(\boldsymbol{\theta}) = y_i - \theta_1(1 - e^{-\theta_2 x_i}),$$

computes gradients via the Jacobian, and updates parameters iteratively using gradient-based optimization.

- (c) Challenges include local minima, sensitivity to initialization, and parameter coupling. These can be addressed with good initial guesses, regularization, and robust optimization algorithms.

Question 7 (*Weighted Least Squares*)

Weighted least squares minimizes $J(\boldsymbol{\theta}) = \frac{1}{2m} \sum_{i=1}^m w_i (y_i - \hat{y}_i(\boldsymbol{\theta}))^2$.

- (a) When and why would you use weighted least squares instead of ordinary least squares? [6 points]
- (b) Derive the weighted normal equations $\mathbf{X}^T \mathbf{W} \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^T \mathbf{W} \mathbf{y}$ where $\mathbf{W}$ is the weight matrix. [8 points]
- (c) How do you choose appropriate weights for different scenarios? Give two examples. [6 points]

Solution.

- (a) Weighted least squares is used when observations have unequal variances or differing reliability, giving more influence to precise measurements.
- (b) The objective can be written as

$$J = (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^T \mathbf{W} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}),$$

leading to

$$\mathbf{X}^T \mathbf{W} \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^T \mathbf{W} \mathbf{y}.$$

- (c) Weights may be chosen as inverse noise variances, or based on confidence levels (e.g., sensor accuracy or repeated measurements).

Question 8 (*Feature Engineering*)

Feature engineering involves transforming input variables to improve model performance.

- (a) How can you use least squares to fit polynomial models like $y = \theta_0 + \theta_1 x + \theta_2 x^2 + \varepsilon$? [5 points]
- (b) What is the design matrix for fitting $y = \theta_0 + \theta_1 \sin(x) + \theta_2 \cos(x) + \varepsilon$? [5 points]
- (c) Explain why these remain "linear" models from a least squares perspective. [4 points]

Solution.

- (a) Polynomial models are handled by augmenting the design matrix with powers of x , allowing standard least squares estimation.

(b) The design matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & \sin(x_1) & \cos(x_1) \\ \vdots & \vdots & \vdots \\ 1 & \sin(x_n) & \cos(x_n) \end{bmatrix}.$$

(c) These models are linear in the parameters $\boldsymbol{\theta}$, even though the basis functions are non-linear in x .